Inconsistency descends along provability: if Gamma proves A and Gamma plus A explodes, Gamma explodes

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: claim_key carries @947: the label also reappears at line 1181 inside a summary proposition's enumerate item, a different statement slot the author reuses.

Editorial note. The label prop:provability-contr names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Reflexivity of derivability. Same part, \ollabel{prop:reflexivity} (source line 836). Line 955: every member of the original Gamma still derives from the extended set.
- Transitivity (cut) of derivability. Same part, \ollabel{prop:transitivity} (source line 856). Line 957: the final collapse Gamma |- falsum from the two lifted derivations.

Source reference. Original proposition prop-reflexivity: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 837–837. Prerequisite pointer invoked at proof line(s) 954; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-transitivity: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 857–857. Prerequisite pointer invoked at proof line(s) 957; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *If* $\Gamma \vdash !A$ *and* $\Gamma \cup \{!A\}$ *is inconsistent, then* Γ *is inconsistent.*

Proof. If $\Gamma \cup \{!A\}$ is inconsistent, then $\Gamma \cup \{!A\} \vdash \bot$. By prop-reflexivity, $\Gamma \vdash !B$ for every $!B \in \Gamma$. Since also $\Gamma \vdash !A$ by hypothesis, $\Gamma \vdash !B$ for every $!B \in \Gamma \cup \{!A\}$. By prop-transitivity, $\Gamma \vdash \bot$, i.e., Γ is inconsistent. $\square$